

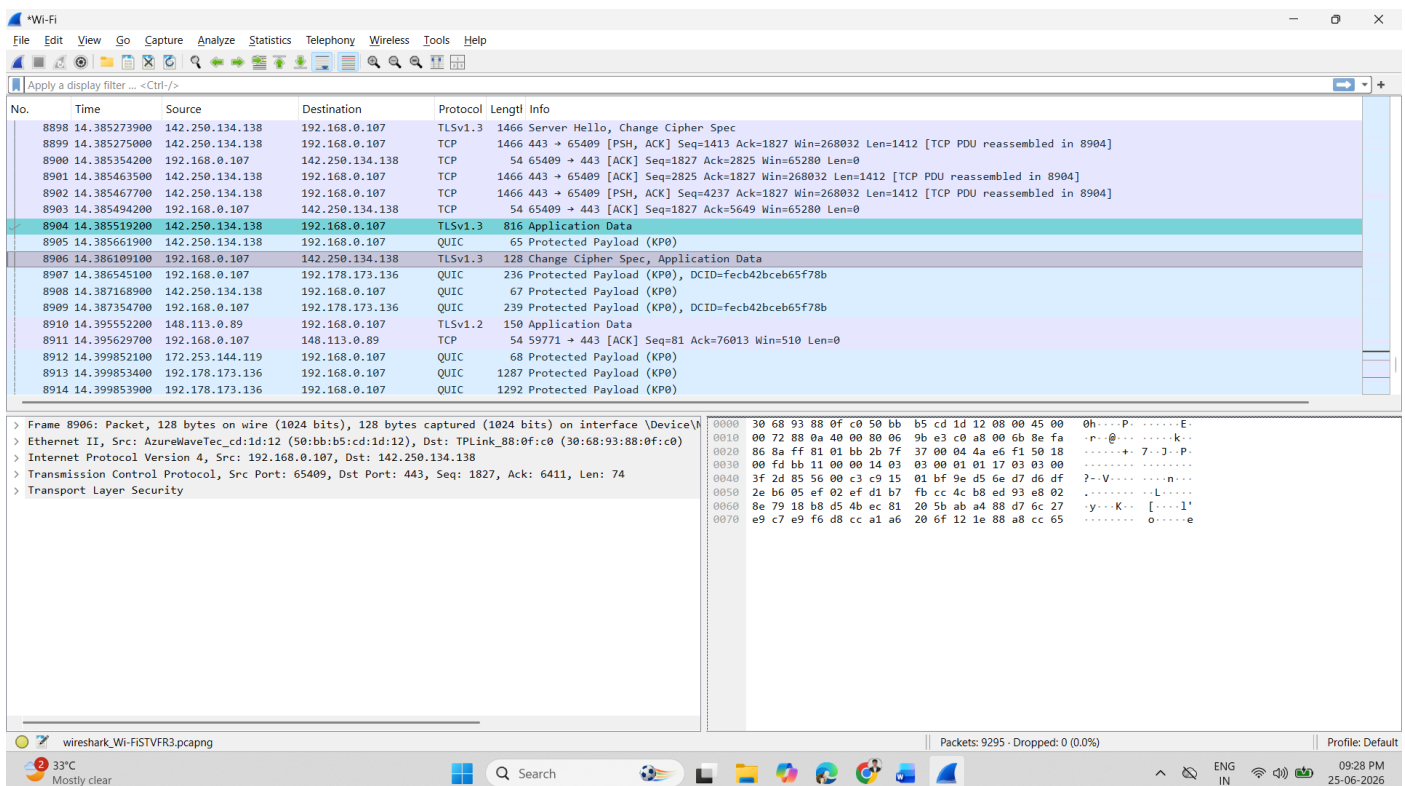
N+ - Network security, Maintenance and Troubleshooting procedures

❖ Software and Hardware Tools

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

ANS :

- -After installing Wireshark and capturing network traffic while browsing Instagram/YouTube for 2 minutes, the following protocols were commonly observed:
 - **TCP (Transmission Control Protocol)** – used for reliable data transfer
 - **DNS (Domain Name System)** – used to resolve domain names into IP addresses
 - **HTTPS (Hypertext Transfer Protocol Secure)** – used for secure web browsing and streaming
- **Top 3 protocols observed:**
 - TCP
 - DNS
 - HTTPS



The screenshot shows the Wireshark interface with a network traffic capture. The packet list on the left shows several packets, with packet 8904 selected. The packet details pane on the right shows the structure of the selected packet, which is a TLSv1.3 Application Data packet. The packet bytes pane at the bottom shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
8898	14.385273900	142.250.134.138	192.168.0.107	TLSv1.3	1466	Server Hello, Change Cipher Spec
8899	14.385275000	142.250.134.138	192.168.0.107	TCP	54	65409 → 65409 [PSH, ACK] Seq=1413 Ack=1827 Win=268032 Len=1412 [TCP PDU reassembled in 8904]
8900	14.385354200	192.168.0.107	142.250.134.138	TCP	54	65409 → 443 [ACK] Seq=1827 Ack=2825 Win=65280 Len=0
8901	14.385463500	142.250.134.138	192.168.0.107	TCP	1466	443 → 65409 [ACK] Seq=2825 Ack=1827 Win=268032 Len=1412 [TCP PDU reassembled in 8904]
8902	14.385467700	142.250.134.138	192.168.0.107	TCP	1466	443 → 65409 [PSH, ACK] Seq=4237 Ack=1827 Win=268032 Len=1412 [TCP PDU reassembled in 8904]
8903	14.385494200	192.168.0.107	142.250.134.138	TCP	54	65409 → 443 [ACK] Seq=1827 Ack=5649 Win=65280 Len=0
8904	14.385519200	142.250.134.138	192.168.0.107	TLSv1.3	816	Application Data
8905	14.385661900	142.250.134.138	192.168.0.107	QUIC	65	Protected Payload (KP0)
8906	14.386109100	192.168.0.107	142.250.134.138	TLSv1.3	128	Change Cipher Spec, Application Data
8907	14.386545100	192.168.0.107	192.178.173.136	QUIC	236	Protected Payload (KP0), DCID=fecb42bceb65f78b
8908	14.387168900	142.250.134.138	192.168.0.107	QUIC	67	Protected Payload (KP0)
8909	14.387354700	192.168.0.107	192.178.173.136	QUIC	239	Protected Payload (KP0), DCID=fecb42bceb65f78b
8910	14.395552200	148.113.0.89	192.168.0.107	TLSv1.2	150	Application Data
8911	14.395629700	192.168.0.107	148.113.0.89	TCP	54	59771 → 443 [ACK] Seq=81 Ack=76013 Win=510 Len=0
8912	14.399852100	172.253.144.119	192.168.0.107	QUIC	68	Protected Payload (KP0)
8913	14.399853400	192.178.173.136	192.168.0.107	QUIC	1287	Protected Payload (KP0)
8914	14.399853900	192.178.173.136	192.168.0.107	QUIC	1292	Protected Payload (KP0)

Frame 8906: Packet, 128 bytes on wire (1024 bits), 128 bytes captured (1024 bits) on interface \Device\NPF...
 Ethernet II, Src: AzureWaveTec, Dst: TPLink, Len: 144
 Internet Protocol Version 4, Src: 192.168.0.107, Dst: 142.250.134.138
 Transmission Control Protocol, Src Port: 65409, Dst Port: 443, Seq: 1827, Ack: 6411, Len: 74
 Transport Layer Security

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

ANS :

- Ping www.spotify.com (Slower)

```
C:\Users\HET>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:19::810] with 32 bytes of data:
Reply from 2a04:4e42:19::810: time=20ms
Reply from 2a04:4e42:19::810: time=18ms
Reply from 2a04:4e42:19::810: time=17ms
Reply from 2a04:4e42:19::810: time=17ms

Ping statistics for 2a04:4e42:19::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 20ms, Average = 18ms
```

- Ping www.zomato.com (Faster)

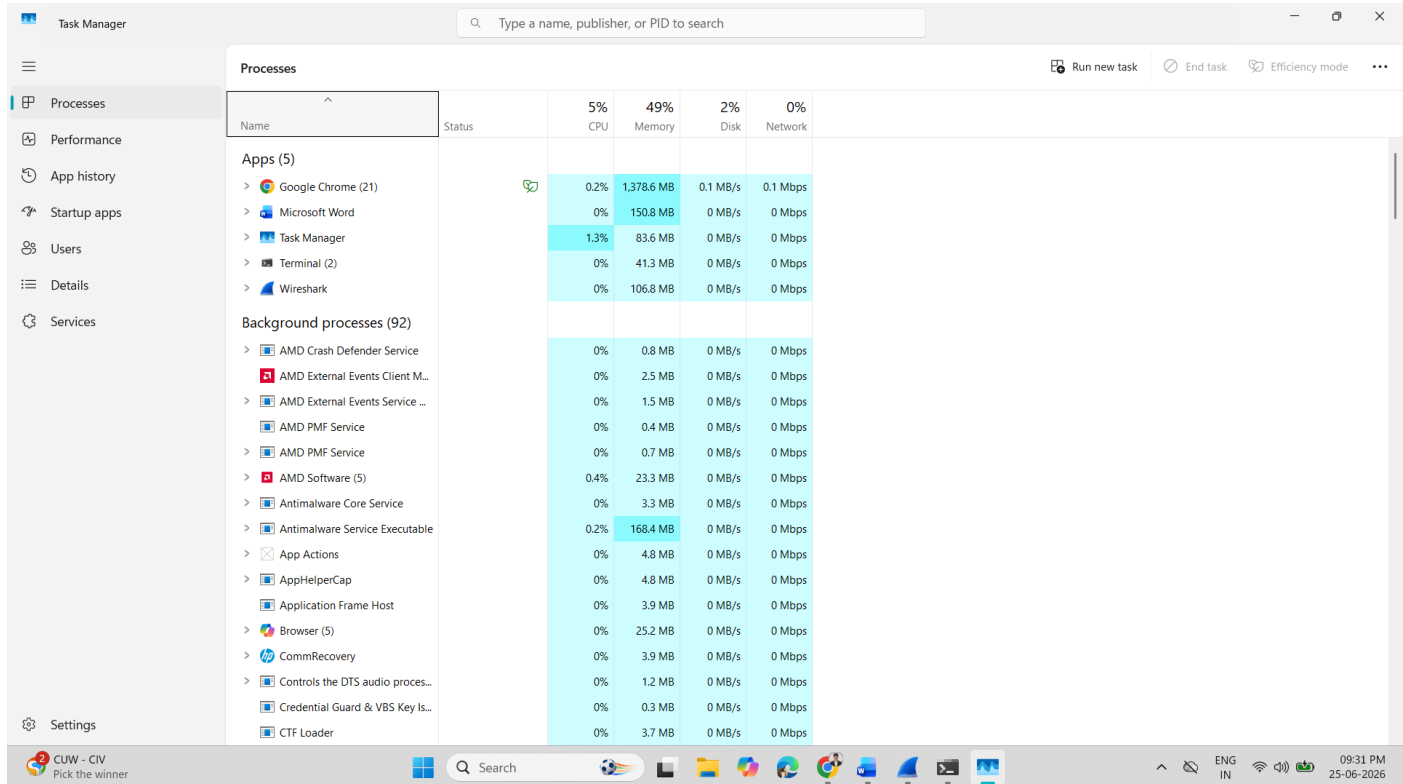
```
C:\Users\HET>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:4::17da:f99a] with 32 bytes of data:
Reply from 2600:140f:4::17da:f99a: time=23ms
Reply from 2600:140f:4::17da:f99a: time=17ms
Reply from 2600:140f:4::17da:f99a: time=15ms
Reply from 2600:140f:4::17da:f99a: time=12ms

Ping statistics for 2600:140f:4::17da:f99a:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 23ms, Average = 16ms
```

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

ANS :

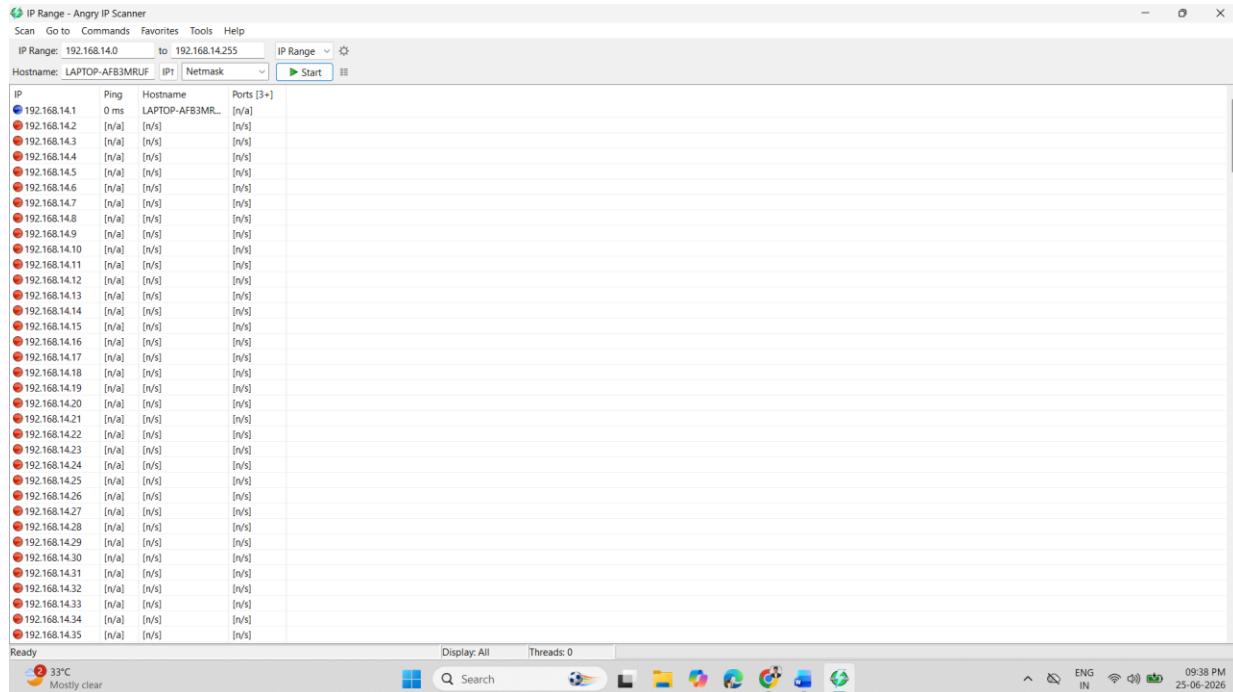


Name	Status	5% CPU	49% Memory	2% Disk	0% Network
Apps (5)					
> Google Chrome (21)		0.2%	1,378.6 MB	0.1 MB/s	0.1 Mbps
> Microsoft Word		0%	150.8 MB	0 MB/s	0 Mbps
> Task Manager		1.3%	83.6 MB	0 MB/s	0 Mbps
> Terminal (2)		0%	41.3 MB	0 MB/s	0 Mbps
> Wireshark		0%	106.8 MB	0 MB/s	0 Mbps
Background processes (92)					
> AMD Crash Defender Service		0%	0.8 MB	0 MB/s	0 Mbps
> AMD External Events Client M...		0%	2.5 MB	0 MB/s	0 Mbps
> AMD External Events Service ...		0%	1.5 MB	0 MB/s	0 Mbps
> AMD PMF Service		0%	0.4 MB	0 MB/s	0 Mbps
> AMD PMF Service		0%	0.7 MB	0 MB/s	0 Mbps
> AMD Software (5)		0.4%	23.3 MB	0 MB/s	0 Mbps
> Antimalware Core Service		0%	3.3 MB	0 MB/s	0 Mbps
> Antimalware Service Executable		0.2%	168.4 MB	0 MB/s	0 Mbps
> App Actions		0%	4.8 MB	0 MB/s	0 Mbps
> AppHelperCap		0%	4.8 MB	0 MB/s	0 Mbps
> Application Frame Host		0%	3.9 MB	0 MB/s	0 Mbps
> Browser (5)		0%	25.2 MB	0 MB/s	0 Mbps
> CommRecovery		0%	3.9 MB	0 MB/s	0 Mbps
> Controls the DTS audio proces...		0%	1.2 MB	0 MB/s	0 Mbps
> Credential Guard & VBS Key Is...		0%	0.3 MB	0 MB/s	0 Mbps
> CTF Loader		0%	3.7 MB	0 MB/s	0 Mbps

Process Name	Application	Memory Usage
chrome.exe	Google Chrome	800 MB – 1500 MB

4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint : Make sure your firewall allows scanning, and only scan your own home/college network

ANS :



IP	Ping	Hostname	Ports [3+]
192.168.14.1	0 ms	LAPTOP-AFB3MRUF	[n/a]
192.168.14.2	[n/a]	[n/a]	[n/a]
192.168.14.3	[n/a]	[n/a]	[n/a]
192.168.14.4	[n/a]	[n/a]	[n/a]
192.168.14.5	[n/a]	[n/a]	[n/a]
192.168.14.6	[n/a]	[n/a]	[n/a]
192.168.14.7	[n/a]	[n/a]	[n/a]
192.168.14.8	[n/a]	[n/a]	[n/a]
192.168.14.9	[n/a]	[n/a]	[n/a]
192.168.14.10	[n/a]	[n/a]	[n/a]
192.168.14.11	[n/a]	[n/a]	[n/a]
192.168.14.12	[n/a]	[n/a]	[n/a]
192.168.14.13	[n/a]	[n/a]	[n/a]
192.168.14.14	[n/a]	[n/a]	[n/a]
192.168.14.15	[n/a]	[n/a]	[n/a]
192.168.14.16	[n/a]	[n/a]	[n/a]
192.168.14.17	[n/a]	[n/a]	[n/a]
192.168.14.18	[n/a]	[n/a]	[n/a]
192.168.14.19	[n/a]	[n/a]	[n/a]
192.168.14.20	[n/a]	[n/a]	[n/a]
192.168.14.21	[n/a]	[n/a]	[n/a]
192.168.14.22	[n/a]	[n/a]	[n/a]
192.168.14.23	[n/a]	[n/a]	[n/a]
192.168.14.24	[n/a]	[n/a]	[n/a]
192.168.14.25	[n/a]	[n/a]	[n/a]
192.168.14.26	[n/a]	[n/a]	[n/a]
192.168.14.27	[n/a]	[n/a]	[n/a]
192.168.14.28	[n/a]	[n/a]	[n/a]
192.168.14.29	[n/a]	[n/a]	[n/a]
192.168.14.30	[n/a]	[n/a]	[n/a]
192.168.14.31	[n/a]	[n/a]	[n/a]
192.168.14.32	[n/a]	[n/a]	[n/a]
192.168.14.33	[n/a]	[n/a]	[n/a]
192.168.14.34	[n/a]	[n/a]	[n/a]
192.168.14.35	[n/a]	[n/a]	[n/a]

- 192.168.14.1 : LAPTOP-AFB3MRUF

5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

ANS:

- **Cable Tester** – Used to check whether Ethernet cables are properly connected or damaged.
- **Crimping Tool** – Used to attach RJ45 connectors to network cables.
- **Multimeter** – Used to measure voltage, continuity, and detect electrical issues in network hardware.